

Paul He

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Education

Nanyang Technological University , PhD in Computer Science <i>Supervisor: Prof. Kwok Yan Lam (College of Computing and Data Science)</i>	Jan 2026 – Dec 2030 Singapore
• PhD student with full research scholarship focusing on reliable NLP University of Toronto , BSc in Computer Science (Honours) <i>Supervisors: Prof. Gerald Penn, Prof. Zhijing Jin</i>	Sep 2021 – May 2025 Toronto, Canada
• Graduated with High Distinction ETH Zürich , Exchange Year in Computer Science • Completed graduate-level coursework in NLP and Machine Learning	Sep 2023 – Aug 2024

Publications

Uncovering Hidden Correctness in LLM Causal Reasoning via Symbolic Verification <i>[EACL 2026 Main – Oral]</i> Paul He, Yinya Huang, Mrinmaya Sachan, Zhijing Jin.	ACL Anthology URL
CCG Revisited: A Multilingual Empirical Study of the Kuhlmann–Satta Algorithm <i>[IWPT 2025]</i> Paul He, Gerald Penn.	ACL Anthology URL
Pointwise Mutual Information as a Performance Gauge for Retrieval-Augmented Generation <i>[NAACL 2025 Main]</i> Tianyu Liu*, Jirui Qi*, Paul He, Arianna Bisazza, Mrinmaya Sachan, Ryan Cotterell. * Equal contribution	ACL Anthology URL

Work Experience

Amazon Web Services (AWS) <i>Applied Scientist II, CloudWatch Logs</i>	Sep 2025 – Dec 2025 Tübingen, Germany
• Conducted open-ended research on reliability and evaluation of large language models in production settings. • Implemented novel evaluation and reliability methods as production-quality code in internal pipelines. • Contributed research directions that developed into ongoing manuscripts and collaborations.	

Research Experience

Causal Reasoning Evaluation for Language Models <i>Supervisor: Prof. Zhijing Jin (Jinesis AI Lab)</i>	Jan 2025 – Aug 2025 Vector Institute
• Proposed symbolic verification for LLM causal reasoning via do-calculus proof search, uncovering 40% additional semantically correct answers missed by surface-level metrics.	
Polynomial-Time CCG Parsing <i>Supervisor: Prof. Gerald Penn</i>	Sep 2024 – Aug 2025 University of Toronto
• Conducted large-scale multilingual experiments identifying empirical complexity thresholds where symbolic parsing outperforms naive baselines.	
Unsupervised Prompt Optimization for RAG using PMI <i>Supervisors: Tianyu Liu, Prof. Ryan Cotterell (Rycolab)</i>	Jun 2024 – Sep 2024 ETH Zürich
• Developed an unsupervised PMI-based document reordering method, improving QA accuracy by up to 4%.	

Preprints & Under Review

Dialogue as Uncertainty Reduction: A Judge-Free Metric for Multi-Turn Dialogue Evaluation <i>[ARR (Under Review)]</i> Paul He, Shiva Kasiviswanathan, Dominik Janzing.
Foundations of Global Consistency Checking with Noisy LLM Oracles <i>[ARR (Under Review)]</i> Paul He, Elke Kirschbaum, Shiva Kasiviswanathan.

Skills

Programming Languages: Python (research & production), C, Java, SQL
Languages: English (fluent), Mandarin (proficient), German (proficient)